

Python HW

There no time limit for this HW.

The coding should be quick. The thnking may take some time.

- In a markdown cell write down the AI model you use if needed
- This time work until you get stuck. If you get stuck DO NOT ask AI for help. Look at your class notebook. Walk away and come back to it later.
- If after thinking about it you are still stuck then yask a friend before asking AI. Give your brain a chance. There are a lot of hints in the class notebook. Don't rush it.
- If you asked AI for help write down what you asked for help on and how long you waited

AI Model:

HW Introduction

The goal of today's homework is get practice reading in files and then thinking about arrays and data

Q1

- Your are only allowed to add two lines of code to answer this question.
- Take your code from today where you read in Brian.csv and plotted the array.
- You can only add two more lines of code to your work from today to make the plot below.
- Using two lines of code and your slicing nomenclature add an exclamation point.
- Remember in the file I had the background as 0's and the writing as 1's.
- You need to at least make a third color or different gray scale!.
- My answer is below. Both answers are perfectly correct. I just wanted to add more color this year! You only need to do 1. I am just showing off. And yes it is still just two extra lines of code.
- To make the lines crisp I turned off interpolation. In imshow you need to add the keyword argument `interpolation='none'` like we did in class.

```
Out[28]: <matplotlib.image.AxesImage at 0x114c74c50>
```



Out[29]: <matplotlib.image.AxesImage at 0x114f59ad0>

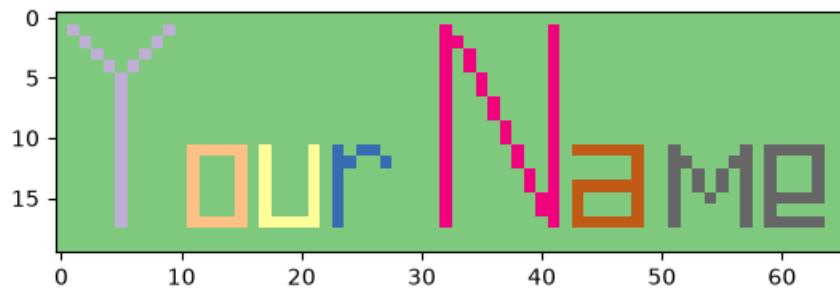


Q2. For part 2.

Make a new csv file in excel and make it so you can print your name in more than one color or gray scale.

- That means Your name must be in multiple colors (so at least 3 colors or gray scales overall). First name, last name, nickname; your choice
- So read in your file and make the plot.
- As a hint. I made my csv file in excel and I made the rows and columns smaller or changed the zoom so I could see the pattern I was entering.
- When saving a csv in excel the ms-dos csv option seems to work the best. When making the csv file name the csv file with your name. Make sure to upload your csv file with your homework.

Out[30]: <matplotlib.image.AxesImage at 0x114ed2210>



Q3

- Make an excel file with one column for x-values and three columns for y-values.
- Label each column.
- You need to do at least ten rows.
- For the y-columns make each distinct. You can choose your own function or pattern (straight, curved, etc) for the y-values.
- But make sure each line is unique.
- Now read in the file and plot the lines and label the axes.
- Summary: make an excel file, read it in, plot the lines
- Make sure to turn in your excel file.

